

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1704

Henry County 230 kV
March 31, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1704. The Interconnection Customer proposes a 1194.1 MW natural gas combined cycle facility interconnecting at the Henry County 230 kV substation in the Indiana area. The request was received on March 27, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 604 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$60,461,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Murdock – HenryCounty 230 KV line
Contingency: East Methuen – Henry County 500 KV
Loading: 166.6% (Rating: 848 MVA)
DFAX: 59.7%
Pre-project loading: 82.6%
2. Facility: Sabanera – Murdock 230 KV line
Contingency: E Methuen – Henry County 500 KV
Loading: 151.4% (Rating: 877 MVA)
DFAX: 53.5%
Pre-project loading: 78.6%
3. Facility: Sabnera Substation 500/230 kV Auto
Contingency: Henry County – E Methuen 500 KV
Loading: 110.9% (Rating: 1,608 MVA)
DFAX: 39.6%
Pre-project loading: 81.5%
4. Facility: Henry County 500/230 kV Transformer
Contingency: Lonetree Substation – E Methuen 500 KV
Loading: 109.8% (Rating: 1,662 MVA)
DFAX: 48.9%
Pre-project loading: 74.7%
5. Facility: Stat St – Waynedale 230 KV
Contingency: Muskogee – Montpelier Electric Generating Sta 500 KV
Loading: 104.5% (Rating: 1,482 MVA)
DFAX: 8.2%
Pre-project loading: 97.9%

6. Facility: Monfort – Waynedale 230 KV line
Contingency: Montpelier Electric Generating Sta – Wabash 500 KV
Loading: 103.8% (Rating: 1,839 MVA)
DFAX: 7.9%
Pre-project loading: 98.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Murdck – Henry County 230 KV line	Rebuild line with higher-capacity conductor	\$24,704,000	78% (\$19,174,000)
Murdock – Sbanera 230 KV	Rebuild line with higher-capacity conductor	\$49,326,000	48% (\$23,836,000)
Sabnera Substation 500/230 kV Auto	Replace transformer with higher-rated unit	\$30,114,000	28% (\$8,272,000)
Henry County 500/230 kV Transformer	Replace transformer with higher-rated unit	\$21,595,000	36% (\$7,885,000)
State St – Waynedale 230 KV	Reconductor line	\$4,950,000	9% (\$442,000)
Monfort – Waynedale 230 KV	Reconductor line	\$10,648,000	8% (\$852,000)
	Total allocated to project		\$60,461,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.